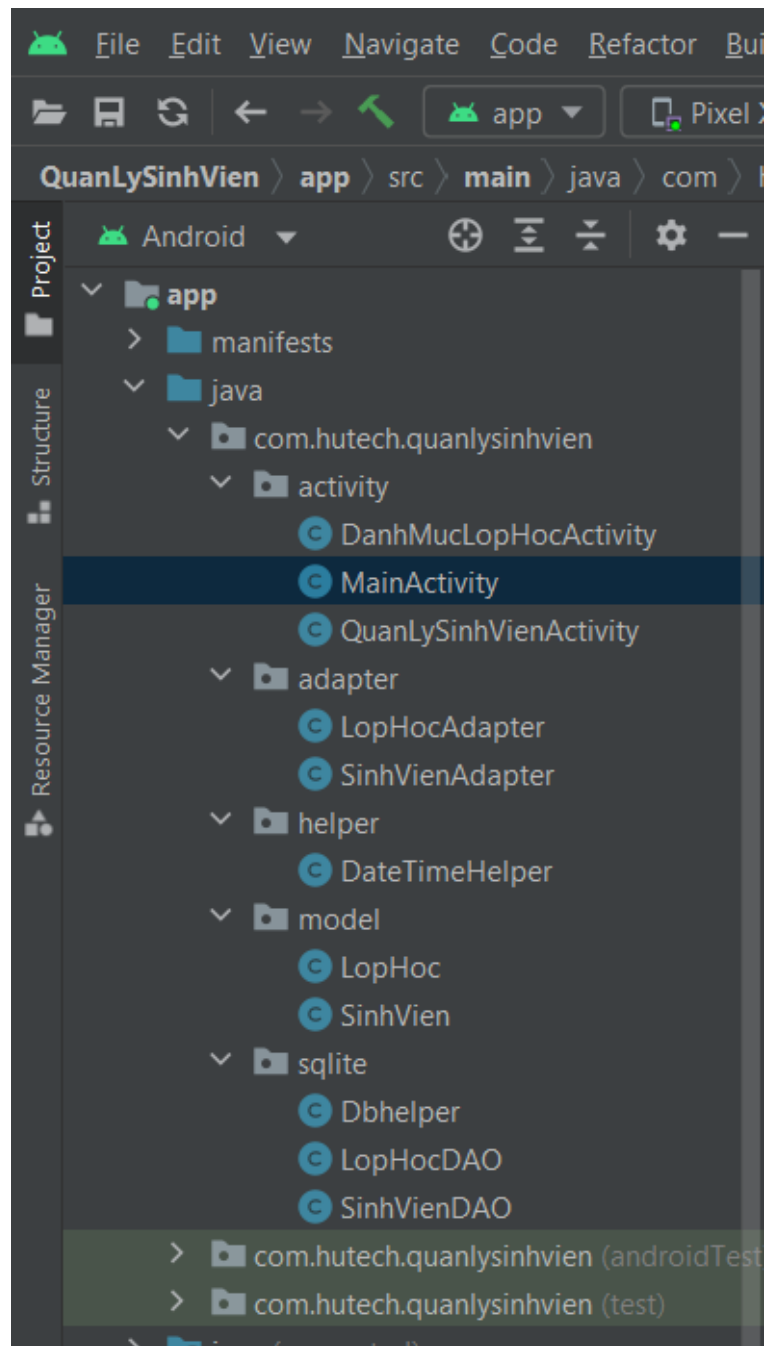






```
public class MainActivity extends AppCompatActivity implements View.OnClickListener
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.activity_main);  
    findViewById(R.id.btnDanhMucLopHoc).setOnClickListener(this);  
    findViewById(R.id.btnQuanLySinhVien).setOnClickListener(this);  
}
```

```
@Override
```

```
public void onClick(View view) {  
    switch (view.getId()){  
        case R.id.btnDanhMucLopHoc:  
            Intent intent = new Intent(MainActivity.this, DanhMucLopHocActivity.class);  
            startActivity(intent);  
            break;  
        case R.id.btnQuanLySinhVien:  
            Intent intent1 = new Intent(MainActivity.this, QuanLySinhVienActivity.class);  
            startActivity(intent1);  
            break;  
    }  
}
```

```
public class DateTimeHelper {  
    private static String Pattern = "dd/MM/yyyy";  
    public static Date toDate(String st) throws ParseException {  
        SimpleDateFormat sdf = new SimpleDateFormat("dd/MM/yyyy");  
        return sdf.parse(st);  
    }  
  
    public static String toString(Date date){  
        SimpleDateFormat sdf = new SimpleDateFormat("dd/MM/yyyy");  
  
        return sdf.format(date);  
    }  
}
```

```
public class Dbhelper extends SQLiteOpenHelper {
    private static final String DB_Name = "QLSinhvien";
    private static final int DB_Version = 1;
    public Dbhelper(@Nullable Context context) {
        super(context, DB_Name, null, DB_Version);
    }
    @Override
    public void onCreate(SQLiteDatabase sqLiteDatabase) {
        String lophocSQL = "CREATE TABLE lophocs(ID INTEGER PRIMARY KEY AUTOINCREMENT, " +
            " tenlop text not null) ";
        String sinhvienSQL = "CREATE TABLE sinhvien(id text primary key, " +
            " hoten text not null, ngaysinh text, lophocid INTEGER, " +
            " FOREIGN KEY (lophocid) REFERENCES lophocs(id))";

        sqLiteDatabase.execSQL(lophocSQL);
        sqLiteDatabase.execSQL(sinhvienSQL);
    }
    @Override
    public void onUpgrade(SQLiteDatabase sqLiteDatabase, int i, int i1) {
        String lophocSQL = "DROP TABLE IF exists lophoc";
        String sinhvienSQL = "DROP TABLE IF EXISTS sinhvien";

        sqLiteDatabase.execSQL(sinhvienSQL);
        sqLiteDatabase.execSQL(lophocSQL);
        onCreate(sqLiteDatabase);
    }
}
```

```
public class Dbhelper extends SQLiteOpenHelper {
    private static final String DB_Name = "QLSinhvien";
    private static final int DB_Version = 1;
    public Dbhelper(@Nullable Context context) {
        super(context, DB_Name, null, DB_Version);
    }
    @Override
    public void onCreate(SQLiteDatabase sqLiteDatabase) {
        String lophocSQL = "CREATE TABLE lophocs(ID INTEGER PRIMARY KEY AUTOINCREMENT, " +
            " tenlop text not null)";
        String sinhvienSQL = "CREATE TABLE sinhvien(id text primary key, " +
            " hoten text not null, ngaysinh text, lophocid INTEGER, " +
            " FOREIGN KEY (lophocid) REFERENCES lophocs(id))";

        sqLiteDatabase.execSQL(lophocSQL);
        sqLiteDatabase.execSQL(sinhvienSQL);
    }
    @Override
    public void onUpgrade(SQLiteDatabase sqLiteDatabase, int i, int i1) {
        String lophocSQL = "DROP TABLE IF EXISTS lophoc";
        String sinhvienSQL = "DROP TABLE IF EXISTS sinhvien";
        sqLiteDatabase.execSQL(sinhvienSQL);
        sqLiteDatabase.execSQL(lophocSQL);
        onCreate(sqLiteDatabase);
    }
}
```

```

public class LopHocDAO {
    private SQLiteDatabase db;
    public LopHocDAO(Context context) {
        Dbhelper dbhelper = new Dbhelper(context);
        this.db = dbhelper.getWritableDatabase();
    }

    public long insert(LopHoc lopHoc){
        ContentValues values = new ContentValues();
        //values.put("id",lopHoc.getId());
        values.put("tenlop",lopHoc.getTenlophoc());
        return db.insert("lophocs",null,values);
    }

    public List<LopHoc> get(String sql, String ... selectArgs){
        List<LopHoc> list = new ArrayList<>();
        Cursor cursor = db.rawQuery(sql,selectArgs);
        while(cursor.moveToNext()){
            LopHoc lopHoc = new LopHoc();
            lopHoc.setId(cursor.getInt(0));
            lopHoc.setTenlophoc(cursor.getString(1));
            list.add(lopHoc);
        }
        return list;
    }
}

```

```

    public List<LopHoc> getAll(){
        String sql = "SELECT * FROM lophocs";
        return get(sql);
    }

    public long delete(int ID){
        return db.delete("lophocs","id=?", new String[] {String.valueOf(ID)});
    }
}

```

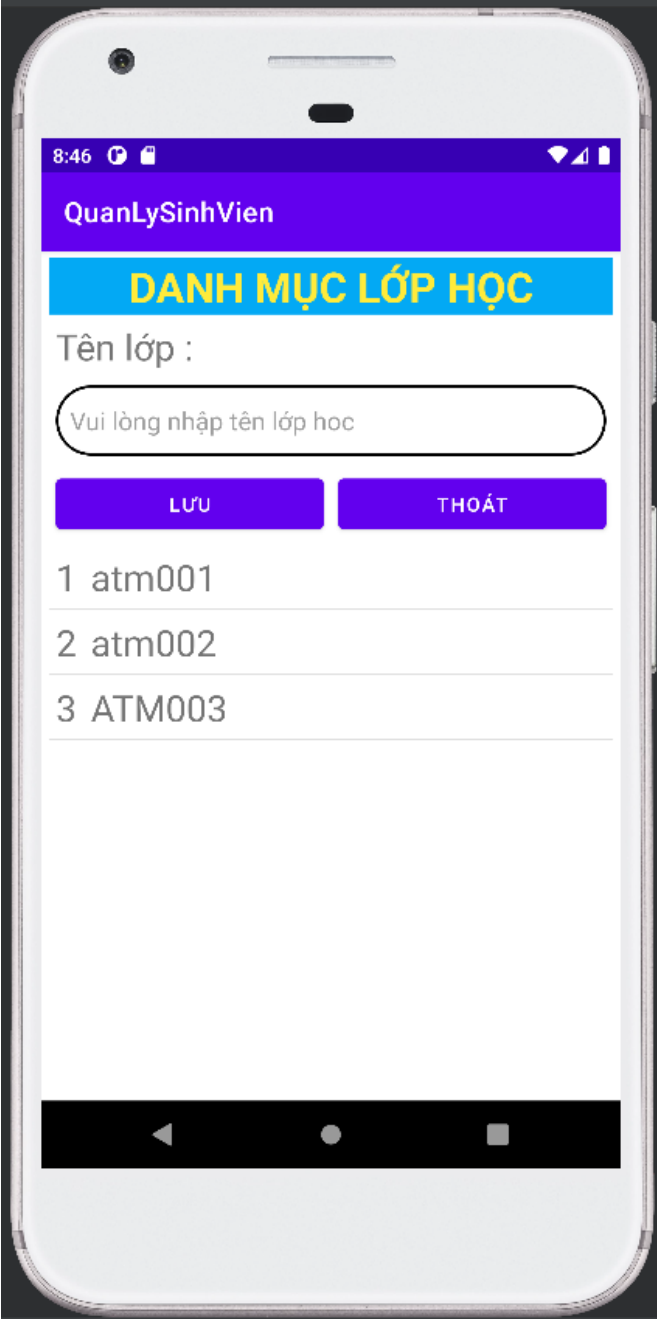
```

public class LopHoc {
    private Integer id;
    private String tenlophoc;
    public LopHoc() {
    }
    public LopHoc(Integer id, String tenlophoc) {
        this.id = id;
        this.tenlophoc = tenlophoc;
    }
    //getter and setter
}

```

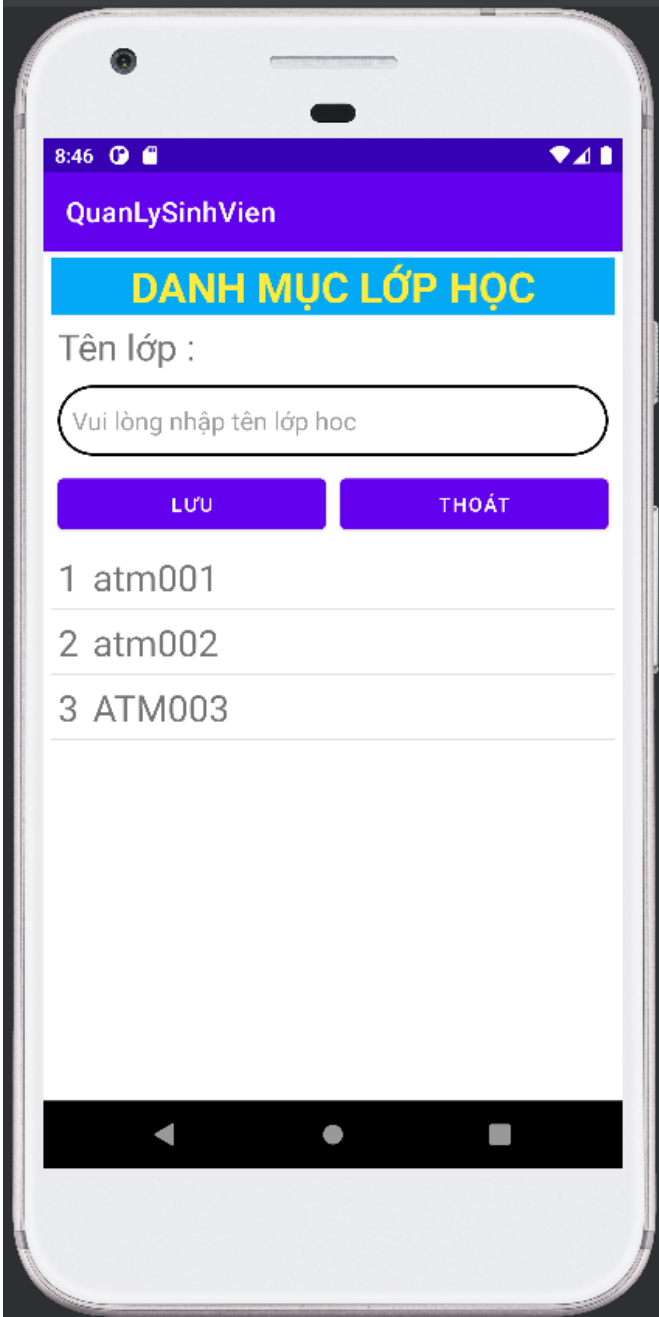
```
public class LopHocAdapter extends BaseAdapter {  
    private Context context;  
    private List<LopHoc> lopHocList;  
    public LopHocAdapter(Context context, List<LopHoc> lopHocList) {  
        this.context = context;  
        this.lopHocList = lopHocList;  
    }  
    @Override  
    public int getCount() {  
        return lopHocList.size();  
    }  
    @Override  
    public Object getItem(int i) {  
        return lopHocList.get(i);  
    }  
    @Override  
    public long getItemId(int i) {  
        return i;  
    }  
}
```

```
@Override  
    public View getView(int i, View view, ViewGroup viewGroup) {  
        if(view == null){  
            view =  
                LayoutInflater.from(context).inflate(R.layout.layout_danh_muc_lop_hoc_items,n  
                ull);  
        }  
        TextView txtIdLopHoc = view.findViewById(R.id.txtIdLopHoc);  
        TextView txtTenLopHoc = view.findViewById(R.id.txtTenLopHoc);  
        LopHoc lopHoc = lopHocList.get(i);  
        txtIdLopHoc.setText("" + lopHoc.getId());  
        txtTenLopHoc.setText(lopHoc.getTenlophoc());  
        return view;  
    }  
}
```



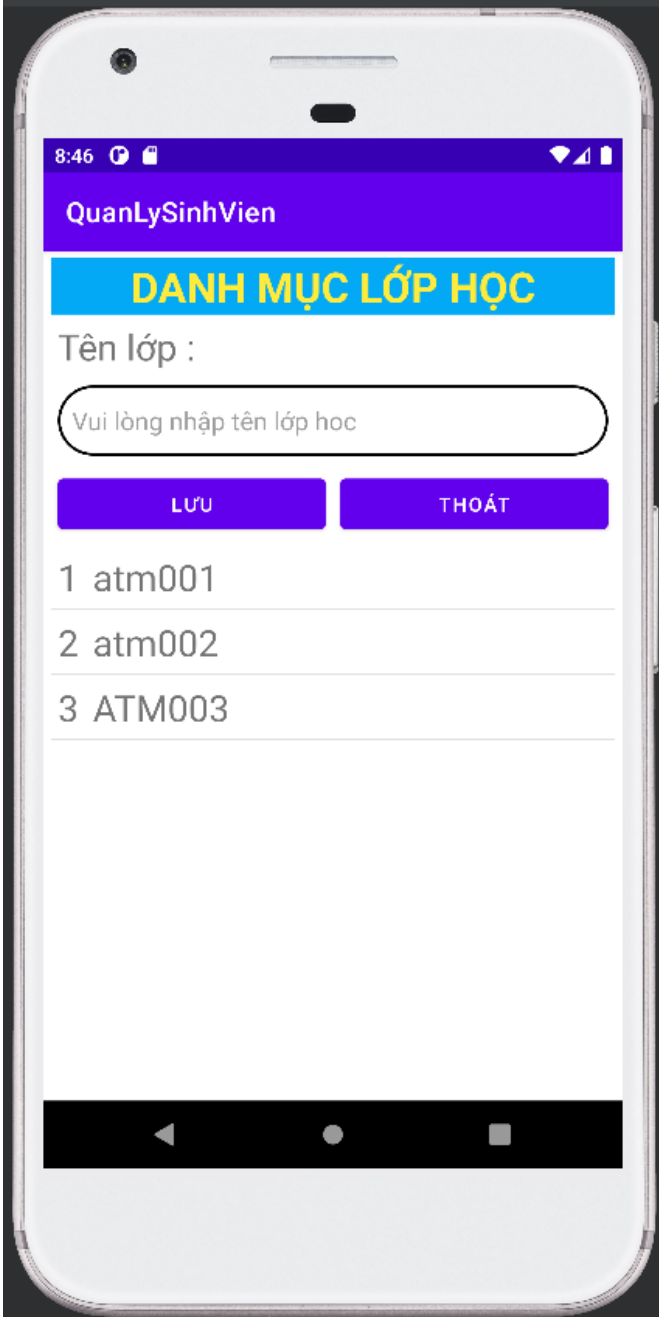
```
public class DanhMucLopHocActivity extends AppCompatActivity implements View.OnClickListener
```

```
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.activity_danh_muc_lop_hoc);  
    findViewById(R.id.btnLuuLopHoc).setOnClickListener(this);  
    findViewById(R.id.btnThoatLopHoc).setOnClickListener(this);  
    edtTenLopHoc = findViewById(R.id.edtTenLopHoc);  
    lvDanhSachLopHoc = findViewById(R.id.lvdanhsachlophoc);  
    fillLopHocListView();  
}
```

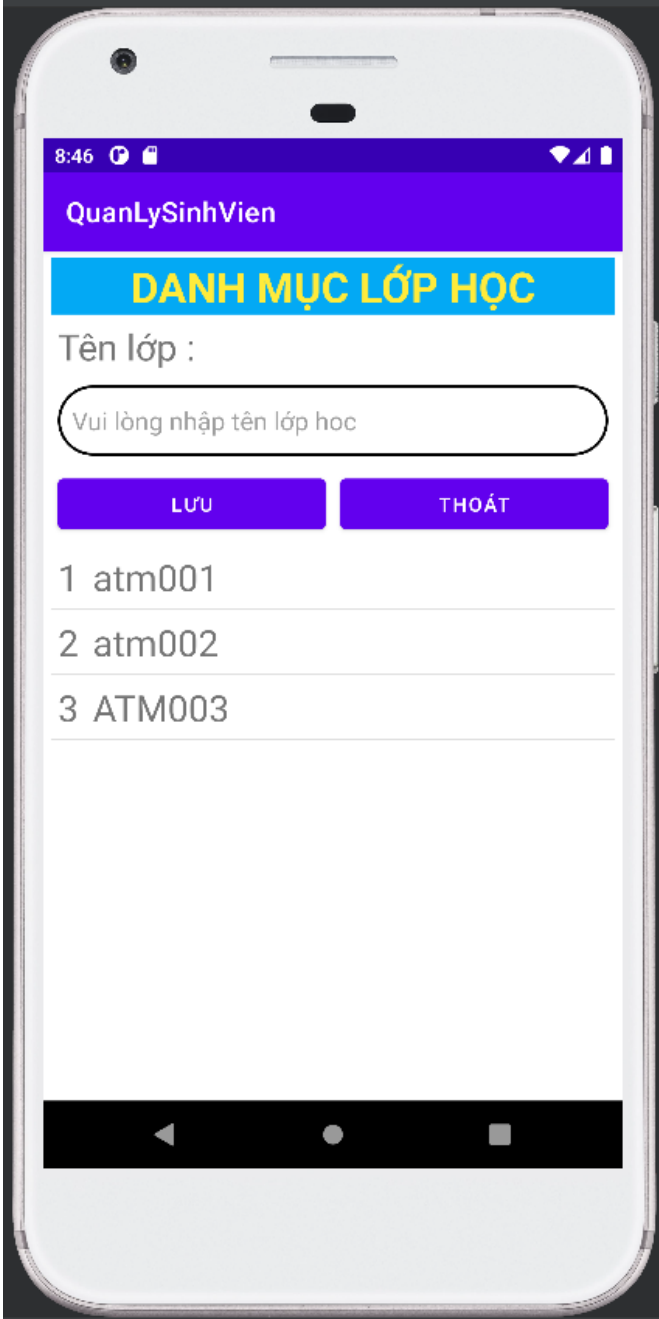



```
private void fillLopHocListView() {
    LopHocDAO lopHocDAO = new LopHocDAO(this);
    lopHocList = lopHocDAO.getAll();

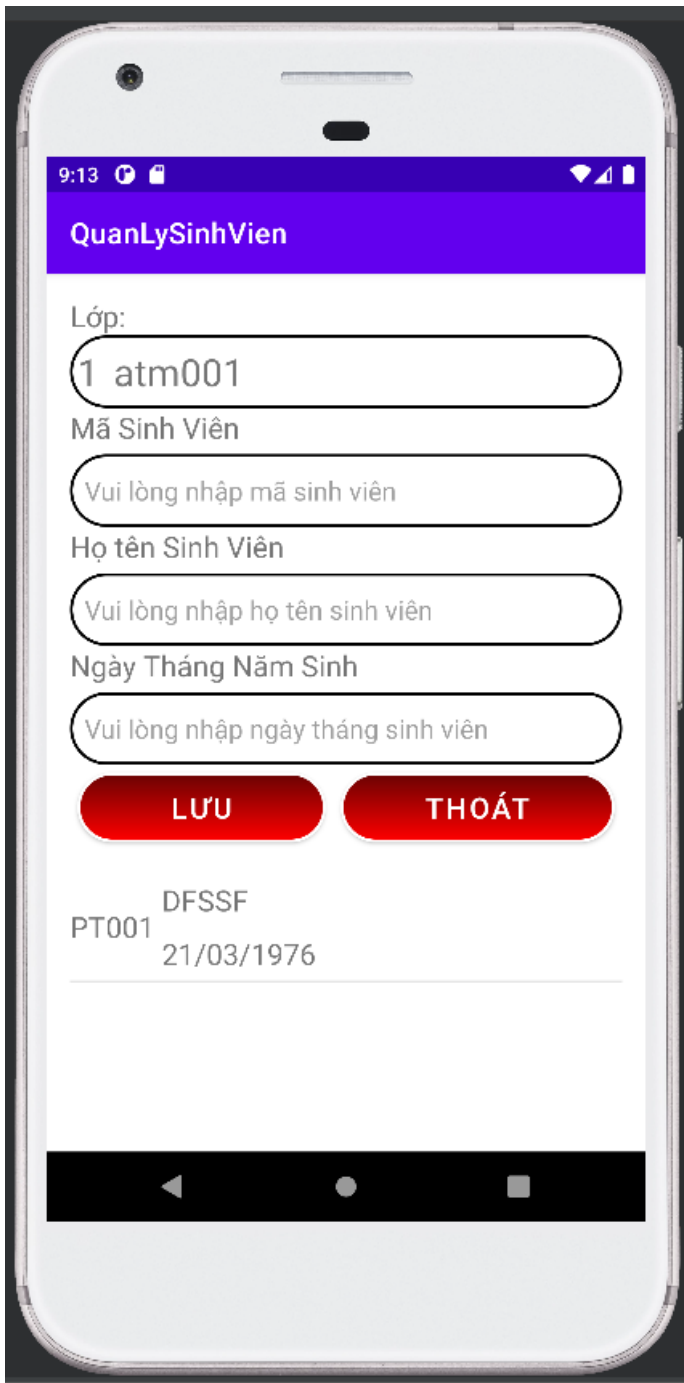
    lopHocAdapter = new LopHocAdapter(this, lopHocList);
    lvDanhSachLopHoc.setAdapter(lopHocAdapter);
    lvDanhSachLopHoc.setOnItemLongClickListener(new AdapterView.OnItemLongClickListener() {
        @Override
        public boolean onItemLongClick(AdapterView<?> adapterView, View view, int i, long l) {
            LopHocDAO lopHocDAO1 = new LopHocDAO(DanhMucLopHocActivity.this);
            LopHoc lopHoc = lopHocList.get(i);
            lopHocDAO.delete(lopHoc.getId());
            fillLopHocListView();
            return false;
        }
    });
}
```



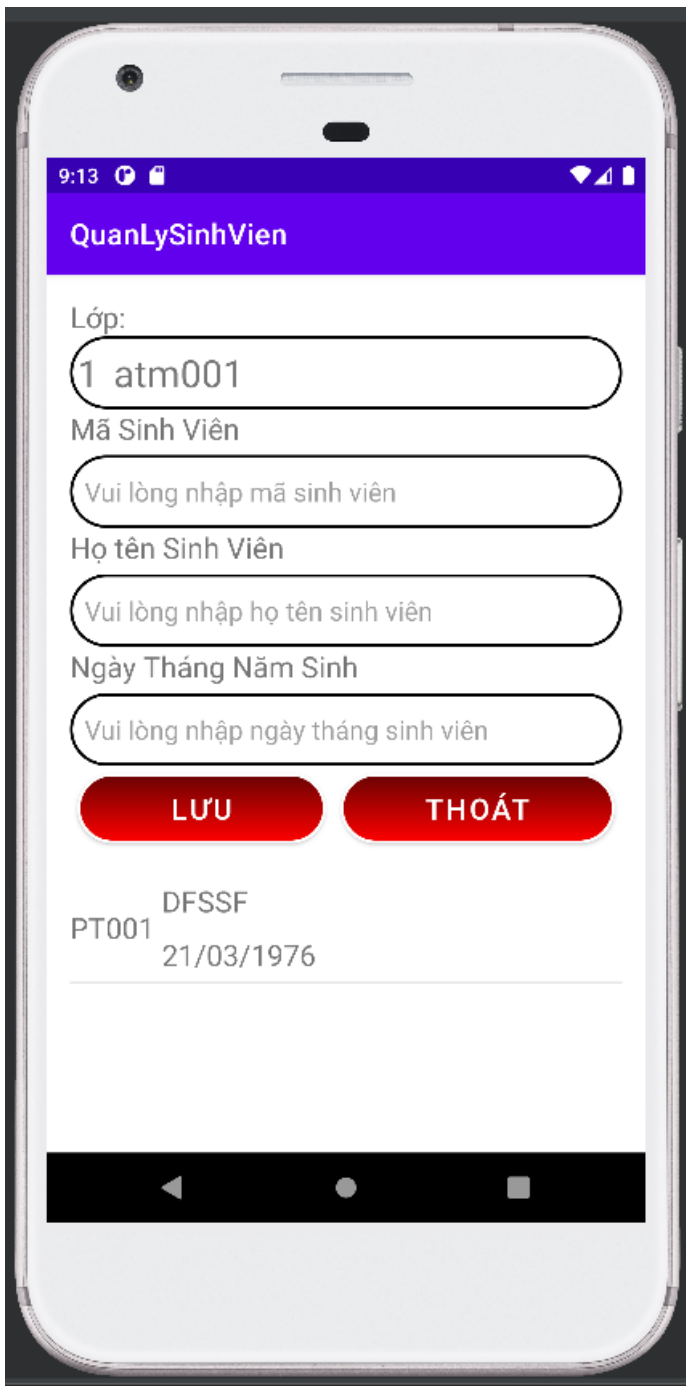
```
public void onClick(View view) {  
    switch (view.getId()){  
        case R.id.btnLuuLopHoc:  
            LopHoc lopHoc = new LopHoc();  
            lopHoc.setTenlophoc(edtTenLopHoc.getText().toString());  
            LopHocDAO lopHocDAO = new LopHocDAO(this);  
            lopHocDAO.insert(lopHoc);  
            fillLopHocListView();  
            Toast.makeText(this, "Đã lưu lớp học", Toast.LENGTH_LONG).show();  
            break;  
        case R.id.btnThoatLopHoc:  
            finish();  
            break;  
    }  
}
```



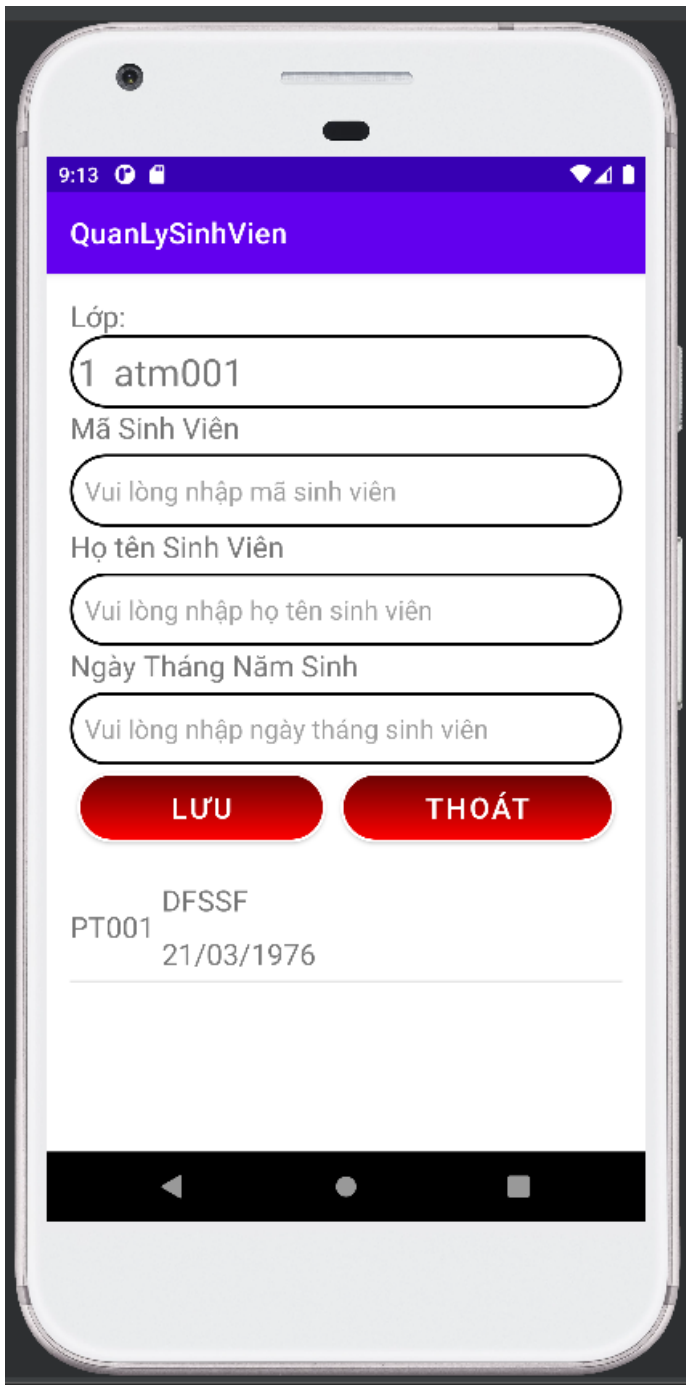
```
public class LopHocAdapter extends BaseAdapter {
    private Context context;
    private List<LopHoc> lopHocList;
    public LopHocAdapter(Context context, List<LopHoc> lopHocList) {
        this.context = context;
        this.lopHocList = lopHocList;
    }
    @Override
    public int getCount() {
        return lopHocList.size();
    }
    @Override
    public Object getItem(int i) {
        return lopHocList.get(i);
    }
    @Override
    public long getItemId(int i) {
        return i;
    }
    @Override
    public View getView(int i, View view, ViewGroup viewGroup) {
        if(view == null){
            view = LayoutInflater.from(context).inflate(R.layout.layout_danh_muc_lop_hoc_items,null);
        }
        TextView txtIdLopHoc = view.findViewById(R.id.txtIdLopHoc);
        TextView txtTenLopHoc = view.findViewById(R.id.txtTenLopHoc);
        LopHoc lopHoc = lopHocList.get(i);
        txtIdLopHoc.setText("" + lopHoc.getId());
        txtTenLopHoc.setText(lopHoc.getTenlophoc());
        return view;
    }
}
```



```
public class SinhVien {  
    private String id;  
    private String hoten;  
    private Date ngaysinh;  
    private Integer lophocid;  
    public SinhVien() {  
    }  
    public SinhVien(String id, String hoten, Date ngaysinh, Integer lophocid) {  
        this.id = id;  
        this.hoten = hoten;  
        this.ngaysinh = ngaysinh;  
        this.lophocid = lophocid;  
    }  
    //getter and setter  
}
```



```
public class SinhVienDAO {
    private SQLiteDatabase db;
    public SinhVienDAO(Context context) {
        Dbhelper dbhelper = new Dbhelper(context);
        this.db = dbhelper.getWritableDatabase();
    }
    public long insert(SinhVien sinhVien){
        ContentValues values = new ContentValues();
        values.put("id",sinhVien.getId());
        values.put("hoten",sinhVien.getHoten());
        values.put("ngaysinh", DateTimeHelper.toString(sinhVien.getNgaysinh()));
        values.put("lophocid",sinhVien.getLophocid());
        return db.insert("sinhvien",null,values);
    }
    @SuppressWarnings("Range")
    public List<SinhVien> get(String sql, String ... selectArgs) throws ParseException {
        List<SinhVien> list = new ArrayList<>();
        Cursor cursor = db.rawQuery(sql,selectArgs);
        while(cursor.moveToNext()){
            SinhVien sinhVien = new SinhVien();
            sinhVien.setId(cursor.getString(cursor.getColumnIndex("id")));
            sinhVien.setHoten(cursor.getString(cursor.getColumnIndex("hoten")));
            sinhVien.setNgaysinh(DateTimeHelper.toDate(cursor.getString(cursor.getColumnIndex("ngaysinh"))));
            sinhVien.setLophocid(cursor.getInt(cursor.getColumnIndex("lophocid")));
            list.add(sinhVien);
        }
        return list;
    }
}
```

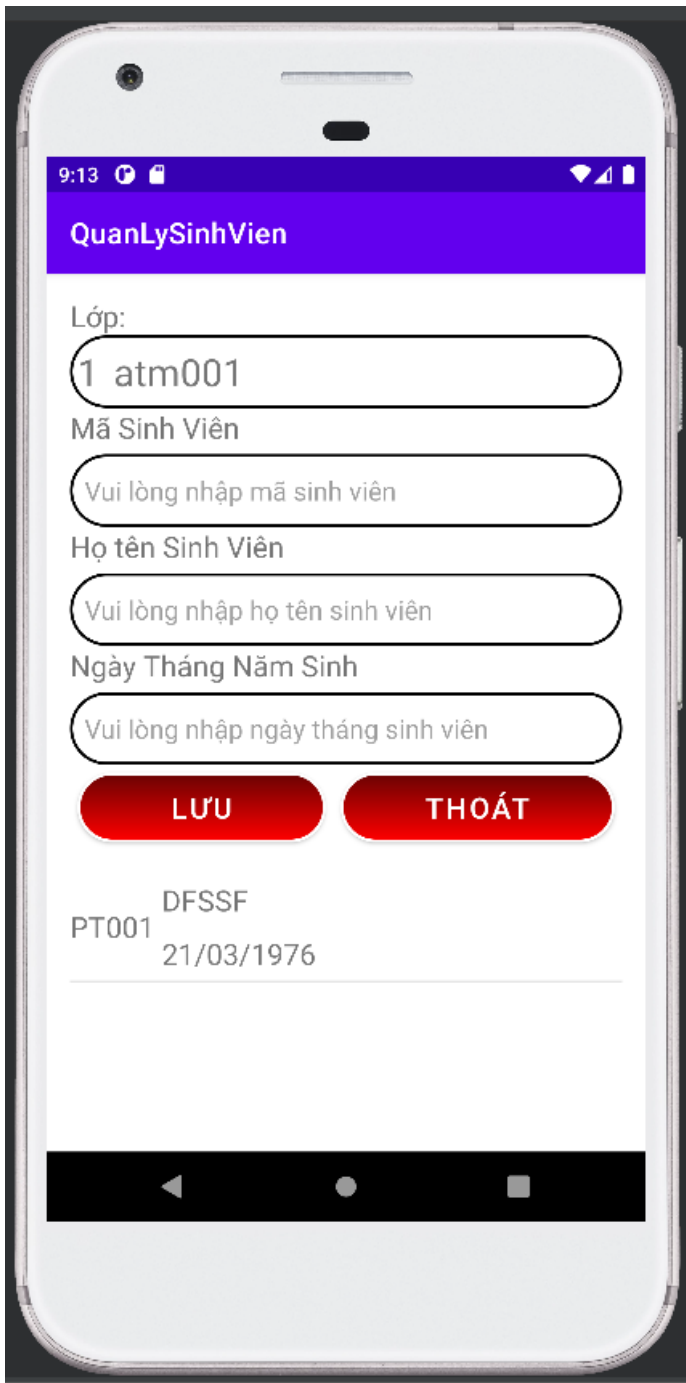


```
public List<SinhVien> getAll() throws ParseException {  
    String sql = "SELECT * FROM sinhvien";  
  
    return get(sql);  
}  
public List<SinhVien> getAllByLophoc(Integer lophocid) throws ParseException {  
    String sql = "SELECT * FROM sinhvien where lophocid = ?";  
    return get(sql, "" + lophocid);  
}  
}
```

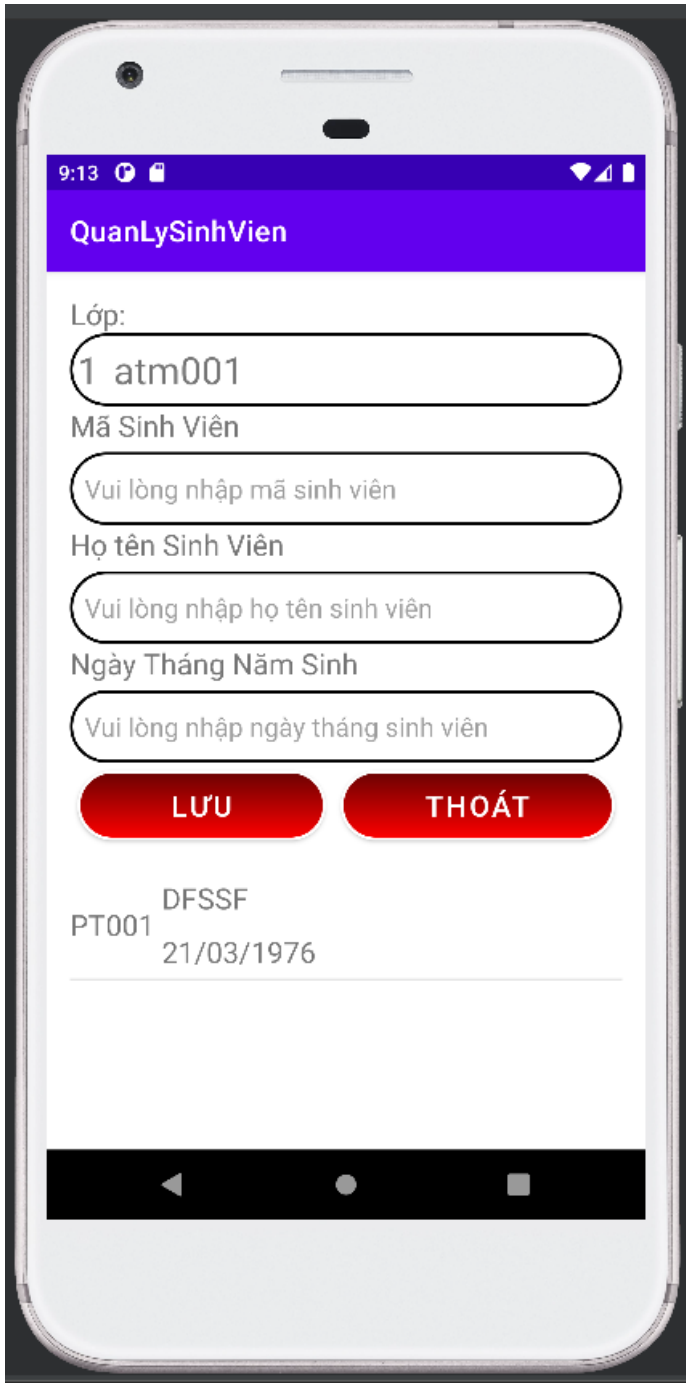
```
public class SinhVienAdapter extends BaseAdapter {  
    private Context context;  
    private List<SinhVien> sinhVienList;  
  
    public SinhVienAdapter(Context context, List<SinhVien> sinhVienList) {  
        this.context = context;  
        this.sinhVienList = sinhVienList;  
    }
```

```
    @Override  
    public int getCount() {  
        return sinhVienList.size();  
    }  
    @Override  
    public Object getItem(int i) {  
        return i;  
    }  
    @Override  
    public long getItemId(int i) {  
        return 0;  
    }  
}
```

```
    @Override  
    public View getView(int i, View view, ViewGroup viewGroup) {  
        if(view == null){  
            view =  
                LayoutInflater.from(context).inflate(R.layout.layout_sinh_vien_items,null);  
        }  
        TextView tvID = view.findViewById(R.id.tvID);  
        TextView tvHoten = view.findViewById(R.id.tvHoten);  
        TextView tvNgaysinh = view.findViewById(R.id.tvNgaysinh);  
        SinhVien sinhVien = sinhVienList.get(i);  
        tvID.setText(sinhVien.getId().toString());  
        tvHoten.setText(sinhVien.getHoten().toString());  
        tvNgaysinh.setText(DateTimeHelper.toString(sinhVien.getNgaysinh()));  
        return view;  
    }  
}
```

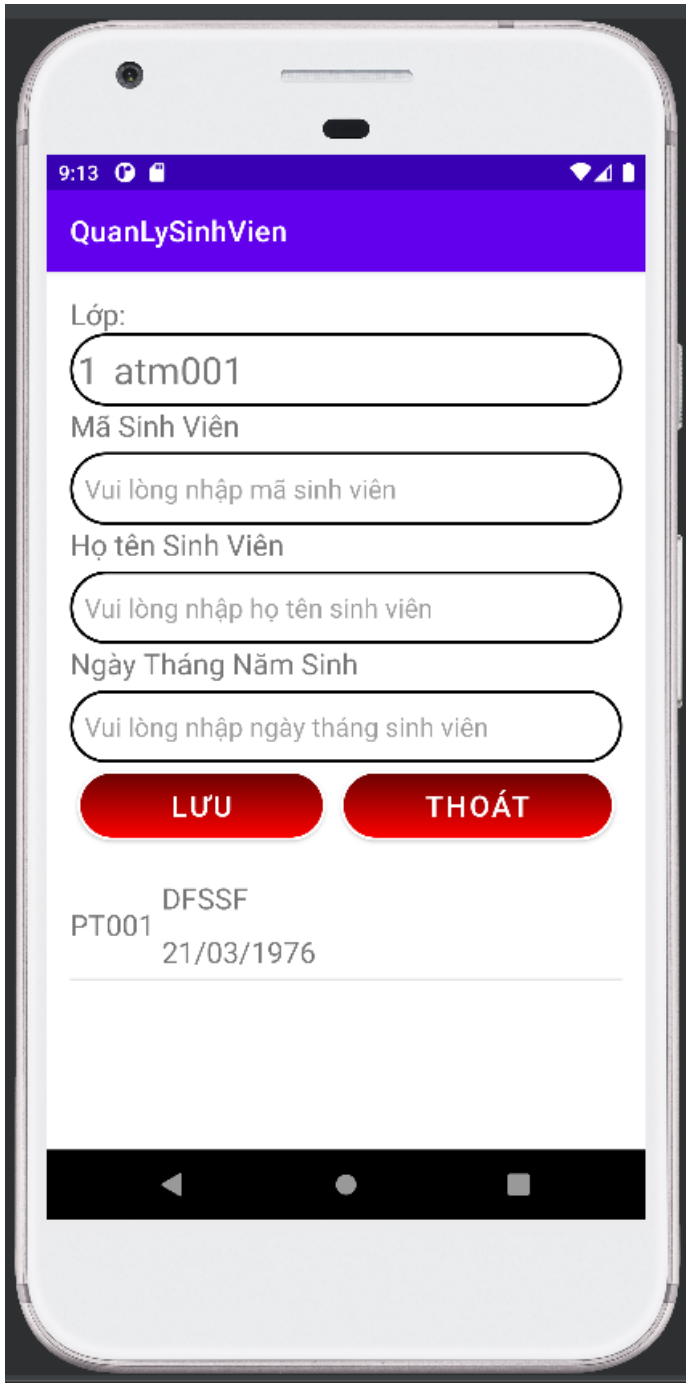


```
EditText edtMaSV, edtHotenSV, edtNgaySinhSV;  
ListView lvDanhSachSinhvien;  
Spinner spLopHoc ;  
SinhVienAdapter sinhVienAdapter;  
private List<LopHoc> lopHocList;  
private List<SinhVien> sinhVienList;  
@Override  
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.activity_quan_ly_sinh_vien);  
    findViewById(R.id.btnLuuSinhVien).setOnClickListener(this);  
    findViewById(R.id.btnThoatSinhVien).setOnClickListener(this);  
    edtMaSV = findViewById(R.id.edtMaSV);  
    edtHotenSV = findViewById(R.id.edtHotenSV);  
    edtNgaySinhSV = findViewById(R.id.edtNgaySinhSV);  
    spLopHoc = findViewById(R.id.spLopHoc);  
    fillLopHocToSpinner();  
    lvDanhSachSinhvien = findViewById(R.id.lvDanhSachSinhvien);  
    fillLopHocToListView();  
}
```

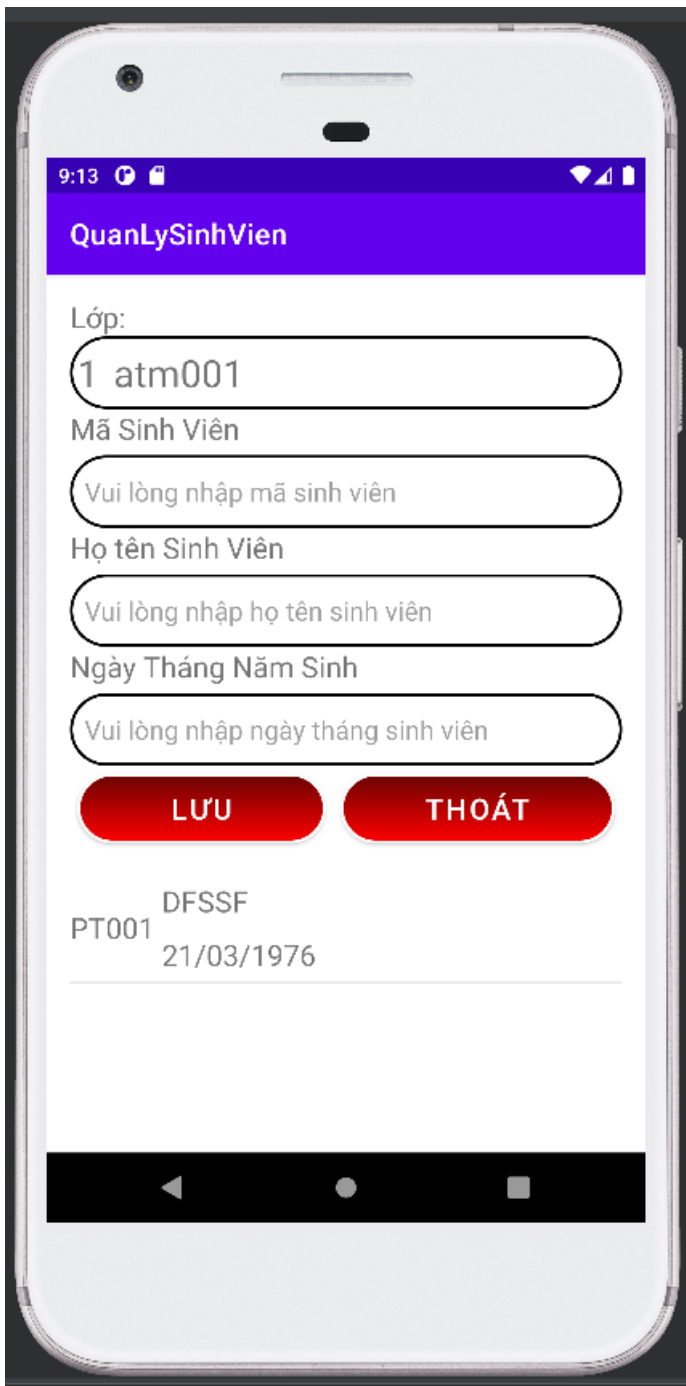
```
spLopHoc.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener() {
    @Override
    public void onItemSelected(AdapterView<?> adapterView, View view, int i, long l) {
        fillLopHocToListView();
    }
    @Override
    public void onNothingSelected(AdapterView<?> adapterView) {
    }
});

private void fillLopHocToListView() {
    SinhVienDAO sinhVienDAO = new SinhVienDAO(this);
    try {
        int lopHocId = lopHocList.get(spLopHoc.getSelectedItemPosition()).getId() ;
        sinhVienList = sinhVienDAO.getAllByLophoc(lopHocId);
        sinhVienAdapter = new SinhVienAdapter(this, sinhVienList);
        lvDanhsachSinhvien.setAdapter(sinhVienAdapter);
    } catch (Exception ex) {
        ex.printStackTrace();
    }
}
```



```
private void fillLopHocToSpinner(){
    LopHocDAO lopHocDAO = new LopHocDAO(this);
    lopHocList = lopHocDAO.getAll();
    LopHocAdapter lopHocAdapter = new LopHocAdapter(this, lopHocList);
    spLopHoc.setAdapter(lopHocAdapter);
}

@Override
public void onClick(View view) {
    SinhVienDAO sinhVienDAO = new SinhVienDAO(this);
    switch (view.getId()){
        case R.id.btnLuuSinhVien:
            try {
                SinhVien sinhVien = new SinhVien();
                sinhVien.setId(edtMaSV.getText().toString());
                sinhVien.setHoten(edtHotenSV.getText().toString());
                sinhVien.setNgaysinh(DateTimeHelper.toDate(edtNgaySinhSV.getText().toString()));
                Toast.makeText(this, "AAA", Toast.LENGTH_LONG).show();
                int lopHoc = spLopHoc.getSelectedItemPosition();
                sinhVien.setLophocid(lopHocList.get(lopHoc).getId());
                Toast.makeText(this, "" + lopHocList.get(lopHoc).getId(), Toast.LENGTH_LONG).show();
                sinhVienDAO.insert(sinhVien);
                Toast.makeText(this, "Sinh viên đã được lưu", Toast.LENGTH_LONG).show();
                fillLopHocToListView();
            } catch (Exception ex) {
                {
                    ex.printStackTrace();
                }
            }
            break;
    }
}
```



```
case R.id.btnThoatSinhVien:
    finish();
    break;
}
}
```